LAST NAME: _____

Practice Quiz 09B — MATH 2055 — Fall 2026

Practice for Quiz 09

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. (a) Construct a function $f : \{1, 2\} \to \{a\}$ and choose subsets $W, X \subseteq \{1, 2\}$ for which
$$f(W \setminus X) \neq f(W) \setminus f(X).$$
Verify your counterexample by computing both sides.

(b) Fix a real $r \neq 1$ and an integer $n \geq 0$. Assume
$$\sum_{i=0}^{n} r^i = \frac{r^{n+1} - 1}{r - 1}.$$
Complete the induction step by proving the corresponding formula for $n + 1$. Explicitly identify the new term.